
SPARSITY AND SUPERPOSITION IN MIXTURE OF EXPERTS

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ABSTRACT

Mixture-of-Experts (MoE) architectures are now a dominant mechanism for scaling large language models, yet their mechanistic divergence from dense networks remains inadequately characterized. Prior analyses establish that dense models exploit *superposition*—the encoding of more features than there are representational dimensions—and that the degree of superposition is a function of feature sparsity and feature importance. We show that this analytical lens does not transfer directly to MoEs. Specifically, we find that neither feature sparsity nor feature importance induces the discontinuous phase transitions observed in dense models, and that a distinct control variable—*network sparsity*, defined as the ratio of active to total experts—more faithfully governs MoE representational behavior. We introduce quantitative metrics for measuring superposition at the granularity of individual experts, and establish empirically that higher network sparsity is associated with greater *monosemanticity*. Building on this, we formalize a definition of expert specialization grounded in monosemantic feature representation rather than in load-balancing statistics, and demonstrate that, under appropriate router initialization, experts converge on coherent feature subsets. Taken together, these results indicate that increasing network sparsity in MoEs can yield more interpretable representations without a corresponding loss in reconstruction performance, contradicting the prevailing assumption that interpretability and capability are inherently in opposition.

1 INTRODUCTION

Mixture-of-Experts (MoE) architectures are now prevalent in state-of-the-art language models, including Qwen3, Mixtral, and Gemini (Yang et al., 2025a; Jiang et al., 2024; Google DeepMind, 2025), chiefly because conditional computation yields favorable compute-to-performance scaling (Shazeer et al., 2017; Fedus et al., 2022). A subsequent line of work refines the routing function (e.g., Expert-Choice routing) and improves optimization stability and transfer (e.g., ST-MoE) (Zhou et al., 2022; Zoph et al., 2022). Complementary theoretical and empirical results establish that a learnable router can recover latent cluster structure in the data distribution, supplying a partial account of why experts specialize (Dikkala et al., 2023). Despite this widespread deployment, however, the internal computation of MoEs remains poorly characterized from a mechanistic-interpretability standpoint.

Several interpretability-oriented methods target the transparency of expert behavior. Yang et al. (2025b) propose MoE-X, which imposes sparsity-aware routing and employs wide, ReLU-activated experts to suppress polysemanticity. Park et al. (2025) introduce MONET, which scales the expert count to support capability editing through direct manipulation of expert activations. These contributions, however, operate primarily at the level of architectural modification; a mechanistic account of how MoEs encode features, how the expert decomposition modulates superposition, and whether experts specialize absent explicit regularization is still lacking. Mu & Lin (2025) survey the MoE literature and identify mechanistic interpretability as a principal open problem.

A central obstacle to interpreting neural networks is the phenomenon of superposition: the regime in which a model encodes strictly more features than it has representational dimensions. Superposition permits a network to pack many sparsely active features into a smaller set of neurons, but this compression renders individual neurons polysemantic and therefore resistant to direct interpretation.

MoE architectures introduce an additional degree of freedom into this problem: network sparsity. Whereas a dense model activates all of its neurons for every input, a MoE activates only a fraction of its total parameters per forward pass, selected by the router (Shazeer et al., 2017). A dense model must therefore exploit feature sparsity by co-locating many sparse features within shared neurons; a MoE, by contrast, can allocate representational capacity more selectively, in principle assigning entire experts to specific feature combinations.

We investigate three questions: (1) whether MoEs exhibit less superposition than their dense counterparts; (2) whether the superposition of a designated feature within a MoE expert undergoes a discrete phase transition as a function of that feature’s relative importance and the overall feature sparsity, as it does in dense models; and (3) whether expert specialization is more precisely characterized through feature representation than through load-balancing statistics.

We address these questions with controlled toy models that extend the framework of Elhage et al. (2022) to the MoE setting. Our principal contributions are as follows: (1) in contrast to dense models, MoEs do not exhibit sharp phase transitions but instead display progressively smoother transitions as network sparsity increases; (2) at matched active- and total-parameter budgets, MoEs consistently attain greater monosemanticity (lower superposition) than dense models, with individual experts encoding features more cleanly; and (3) we formalize an interpretability-oriented definition of expert specialization in terms of monosemantic feature representation and show that experts organize around coherent feature combinations rather than arbitrary load-balancing assignments.

2 RELATED WORK

Superposition and Feature Representations. The Linear Representation Hypothesis posits that networks encode concepts as directions in activation space (Park et al., 2024), yet the count of interpretable features routinely exceeds the ambient dimensionality. Elhage et al. (2022) formalized this regime as superposition, showing that dense models employ non-orthogonal feature packing to maximize encoded capacity at the cost of polysemanticity. Although superposition is provably efficient in capacity terms (Scherlis et al., 2025), it requires post-hoc disentanglement procedures—most notably Sparse Autoencoders—to recover monosemantic features (Bricken et al., 2023). Subsequent work characterizes how feature correlations in the data shape superposition (Prieto et al., 2025) and how interference patterns arise and can be attenuated (Gurnee et al., 2023). We extend this line of analysis to MoE architectures and show that network sparsity, rather than feature sparsity in isolation, is the governing determinant of the representational strategy.

Interpretability of MoEs. Although MoEs constitute the standard mechanism for scaling large language models (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024), mechanistic understanding of their internal representations trails that of dense networks. Existing analyses concentrate on macroscopic quantities—routing stability (Zoph et al., 2022), expert-choice statistics (Zhou et al., 2022), and latent cluster discovery (Dikkala et al., 2023). More recent interpretability-oriented methods attempt to enforce specialization through explicit architectural constraints, such as predefined concept routing (Yang et al., 2025b) or scaling the expert count to the vocabulary size (Park et al., 2025). Such methods, however, generally prioritize capability editing over an explanation of the intrinsic feature geometry. We address this gap by quantifying how the expert decomposition modulates superposition, establishing that MoEs attain greater monosemanticity than dense models, and formalizing a feature-based definition of expert specialization.

3 BACKGROUND

A central objective of mechanistic interpretability is to reverse-engineer a trained network by decomposing its representations into a set of human-interpretable concepts, termed ‘features’. Under the standard linearity assumption, any hidden state h admits the decomposition

$$h = \sum_{i \in F} \alpha_i \vec{f}_i + \vec{b}$$

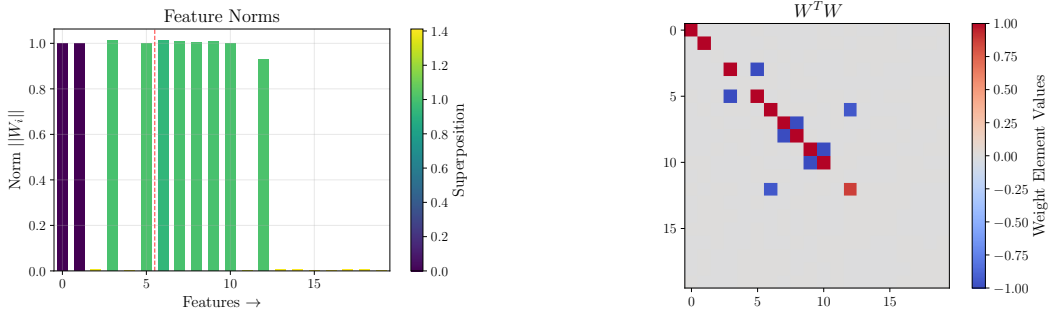
where $\vec{f}_i$ denotes the direction associated with feature i , the scalar coefficient α_i denotes the activation strength of that feature (informally, the degree to which feature i is present in the input), and F denotes the set of all represented features.

The *superposition hypothesis* asserts that a model can represent substantially more features than it has dimensions, i.e., $|F| > m$ for $h \in \mathbb{R}^m$ (Elhage et al., 2022). Encoding more features than dimensions within a latent space requires that the feature vectors in F be packed non-orthogonally. When two features exhibit non-zero interference, $\langle \vec{f}_i, \vec{f}_j \rangle \neq 0$, we say they are in *superposition*. Such superposition is tolerable provided the features are sufficiently sparsely active—an empirical property of most natural-language corpora—but it degrades interpretability, since the recovered coefficient α_i then carries spurious contributions unrelated to the presence of feature i in the input.

Monosemanticity is a property of individual neurons: a neuron is monosemantic when its activation corresponds cleanly to a single coefficient α_i , i.e., when features are basis-aligned. When features are orthogonal (hence not in superposition) but not basis-aligned, the neurons remain *polysemantic* even though inter-feature interference is negligible. This work targets the reduction of superposition rather than the enforcement of basis alignment. For brevity, when we characterize a model, expert, or feature as “more monosemantic,” we mean specifically that it exhibits less superposition.

4 DEMONSTRATING SUPERPOSITION

MoEs are frequently conceptualized as compositions of dense models, with each expert treated as an independent dense subnetwork. Whether experts in fact encode features in the same manner as dense models, however, is not established a priori. We test this hypothesis by directly comparing the superposition induced by MoE and dense architectures under matched conditions.



(a) Per-feature weight-vector norm $\|W_i\|$; color encodes superposition status (green: feature in superposition; purple: monosemantic feature).

(b) Gram matrix $W^T W$, whose (i, j) entry is $(\hat{W}_i \cdot W_j)$, exposing the pairwise interference structure across features.

Figure 1: Feature representation and superposition in a dense model with $n = 20$ features and $m = 6$ hidden dimensions, importance $I = 0.7^i$, and uniform feature density $(1 - S) = 0.1$. The per-feature superposition score (color) is $\sum_j (\hat{W}_i \cdot W_j)^2$.

4.1 EXPERIMENTAL SETUP

Our objective is to characterize how a MoE compresses a high-dimensional input $x \in \mathbb{R}^n$ into lower-dimensional per-expert representations $h \in \mathbb{R}^m$ and subsequently reconstructs it. The construction extends the toy-model framework of Elhage et al. (2022) to the MoE setting.

Input distribution. The input vector x models the activations of an idealized, fully disentangled reference model in which each coordinate x_i corresponds to a distinct, statistically independent feature—that is, a perfectly neuron-aligned, monosemantic representation. We sample $x_i \sim U(0, 1)$ subject to a sparsity parameter $S \in [0, 1]$, under which each coordinate is zeroed independently with probability $P(x_i = 0) = S$. Because features contribute unequally to the objective, we assign each feature x_i a scalar importance I_i that reweights its reconstruction error. To isolate and probe phase transitions for a single feature, we further scale the magnitude of the last feature by a factor $r \in \mathbb{R}^+$, yielding $I = (1, 1, \dots, r)$. The parameters I_i and r thus modulate per-feature signal strength, enabling controlled measurement of expert sensitivity to feature magnitude.

Model architecture. The MoE comprises E experts; each expert e is parameterized by a weight matrix $W^e \in \mathbb{R}^{m \times n}$ and a bias $b^e \in \mathbb{R}^n$. A learned router $g(x) = \text{softmax}(W^r x)$, with $W^r \in \mathbb{R}^{E \times n}$, assigns each input to its top- k experts. Each active expert projects the input to a lower-dimensional hidden state $h^e = W^e x$ and emits the reconstruction $\hat{x}^e = \text{ReLU}((W^e)^\top h^e + b^e)$ ¹. The final output is the gating-weighted sum over active experts, $x' = \sum_{e \in \text{top-}k} w_e \hat{x}^e$, where the w_e are the renormalized gating weights.

Models are trained to minimize an importance-weighted L_2 reconstruction objective, $\mathcal{L} = \sum_x \sum_i I_i (x_i - x'_i)^2$. In Section 4, to prevent expert collapse, we augment this objective with the standard auxiliary load-balancing term (Fedus et al., 2022), $\mathcal{L}_{\text{aux}} = \alpha N \sum_{e=1}^N f_e P_e$, where N is the total number of experts, f_e is the fraction of batch samples routed to expert e , P_e is the mean gating probability assigned to expert e over the batch, and the coefficient $\alpha = 0.01$ controls the penalty strength. We deliberately omit the auxiliary load-balancing term in Section 4 in order to isolate the intrinsic architectural bias of the MoE with respect to superposition.

4.2 MEASURING FEATURE CAPACITY

To compare feature representations across architectures, we quantify two fundamental properties of each feature: its *representation strength* and its *interference* with other features. Representation strength within expert e is measured by the weight-vector norm $\|W_i^e\|$, which indexes the degree to which feature i is encoded by that expert: $\|W_i^e\| \approx 1$ indicates that feature i is fully represented in expert e , whereas $\|W_i^e\| \approx 0$ indicates that it is not learned. Interference between feature i and the remaining features within expert e is visualized via the Gram matrix $W^\top W$, whose off-diagonal entries quantify pairwise interference.

As shown in Figures 1a and 2a, at equal total parameters ($m_{\text{dense}} = \sum m_{\text{experts}} = 6$) the dense and MoE models represent a comparable number of features (10 versus 8) with similar norms. Although the MoE experts exhibit residual local superposition (e.g., Expert 0 in Fig. 2b), the global interference structure is strictly partitioned. Whereas in the dense model (Figure 1b) any feature may interfere with any other, the MoE imposes a block-diagonal interference structure in which features routed to distinct experts have identically zero interference. This indicates that MoEs allocate representational capacity by partitioning the feature space, thereby restricting the global scope of interference.

Expert feature dimensionality. To characterize how the MoE allocates its finite representational capacity relative to the dense model, we measure feature dimensionality, which

¹The output ReLU enforces non-negative reconstructions, consistent with the input distribution in which $x_i \in [0, 1]$ whenever nonzero. In addition, the off-diagonal terms of $(W^e)^\top W^e$ induce negative interference, which the ReLU clips to zero. Under sparse activation this negative interference is effectively cost-free, since it is suppressed at the output, which incentivizes configurations with negative off-diagonal terms (e.g., antipodal feature pairs).

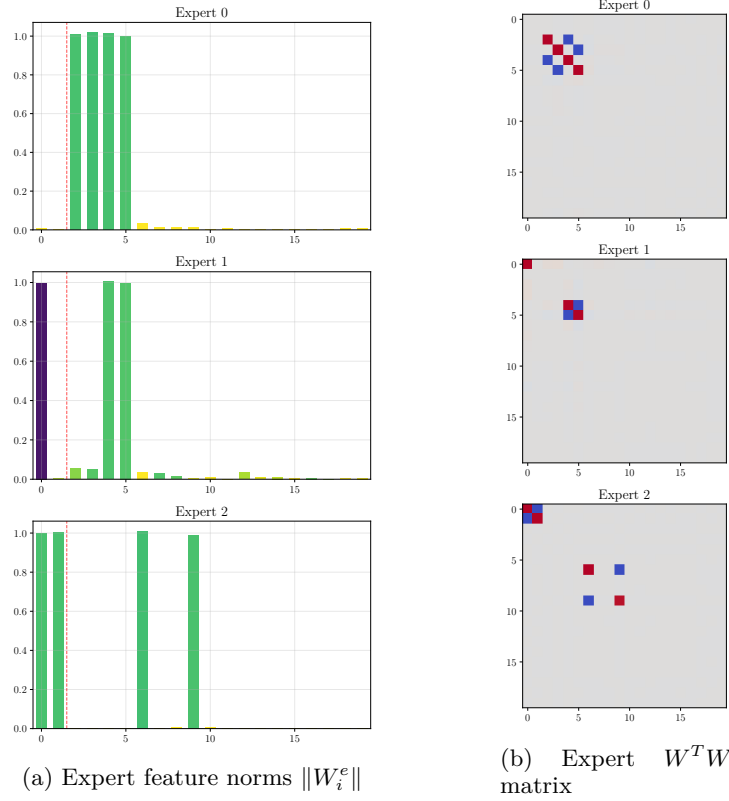


Figure 2: Feature representation and superposition in a MoE with $n = 20$ features, 3 total experts, $m = 2$ hidden dimensions per expert, and top- $k = 1$ routing, under importance $I = 0.7^i$ and feature density $1 - S = 0.1$.

quantifies the “fraction of a dimension” apportioned to a given feature (Elhage et al., 2022). For a feature i , we define its dimensionality within expert e as

$$D_i^e = \frac{\|W_i^e\|^2}{\sum_j (\hat{W}_i^e \cdot W_j^e)^2} \quad (1)$$

The quantity D_i^e is bounded in $[0, 1]$, with 0 indicating an unlearned feature and 1 indicating a monosemantic one. The aggregate capacity of a MoE is correspondingly defined as $D = \sum_e \sum_{i=1}^n D_i^e$.

Efficient packing. When features are “efficiently packed” within a model’s representation space, their dimensionalities sum to the number of embedding dimensions, $\sum_{i=1}^n D_i^e \approx m$ (Cohen et al., 2014; Scherlis et al., 2025; Elhage et al., 2022). For a MoE the analogous relation is $\sum_e \sum_{i=1}^n D_i^e \approx E \cdot m$. Empirically, both dense and MoE models satisfy this dimensionality constraint, establishing that, at equal total parameters, MoEs pack features with the same efficiency as dense models.

Features per dimension and network sparsity. Because both dense and MoE models “efficiently pack” features in their representation space, we compare them via the number of features per dimension. This statistic provides an exact measure of superposition in each architecture and of how the MoE expert count modulates superposition across feature-sparsity levels. A features-per-dimension value exceeding one indicates superposition, since the model then represents more features than it has dimensions. For a MoE we define features per dimension as

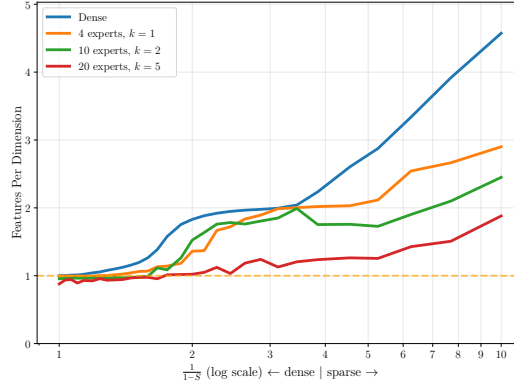


Figure 3: Features per dimension as a function of inverse feature density ($\frac{1}{1-S}$) for **dense** and **MoE** architectures under uniform feature importance ($I_i = 1.0$). The dense model ($n = 100$, $m = 20$) exhibits the greatest superposition; superposition decreases monotonically with expert count: 4 experts, $m = 5$, $k = 1$ (orange); 10 experts, $m = 2$, $k = 2$ (green); 20 experts, $m = 1$, $k = 5$ (red). All configurations hold total parameters fixed and k/E approximately constant. The dashed line at 1.0 marks the monosemantic limit.

$$\frac{1}{k} \sum_{e=1}^E p_e \frac{\|W^e\|_F^2}{m} \quad (2)$$

where $\|W\|_F^2$ denotes the squared Frobenius norm and p_e denotes the expected utilization of expert e over a batch of inputs, i.e., the mean renormalized gating weight after top- k routing.

At *equal* total parameters, the dense model attains a strictly higher features-per-dimension value than the MoE (Figure 3), i.e., greater superposition. The dense model therefore exploits superposition to represent a larger total number of features ($n_{\text{learned}} > m_{\text{total}}$), whereas the MoE saturates near the monosemantic limit ($n_{\text{learned}} \approx m_{\text{total}}$). Moreover, increasing the total expert count—while holding total parameters and the ratio k/E approximately fixed—monotonically decreases features per dimension and hence superposition: *the greater the number of experts, the less superposition in the model*, i.e., features become progressively more monosemantic as experts are added. The additional superposition available to the dense model also confers consistently lower reconstruction loss than the MoEs, with a loss gap of ~ 0.03 – 0.08 at any fixed sparsity (Figure 6, Appendix A.1); this gap narrows as the expert count grows, and the MoEs attain loss comparable to the dense model. Appendix A.2 provides a theoretical account of this behavior.

5 PHASE CHANGE

Although MoEs and dense models learn a comparable number of features, MoEs distribute those features across experts with substantially reduced interference. This suggests that network sparsity reshapes how features are allocated rather than how many are learned. We therefore examine how properties of the input distribution—specifically feature sparsity and importance—drive this allocation, and whether they induce the physics-inspired *phase transitions* previously reported for dense representations.

A model has only a finite set of representational outcomes for any feature: it may ignore the feature, place it in superposition, or represent it monosemantically. A phase transition is the empirical observation that the boundaries between these regimes are, under some conditions, discrete functions of feature sparsity and relative importance.

Dense toy models exhibit discontinuous ‘phase transitions’ between internal feature representations (Elhage et al., 2022). Varying the sparsity and relative importance of the input features elicits distinct regimes; for instance, greater feature sparsity promotes greater superposition. Analyzing the per-expert phase diagram of a MoE reveals that its experts adopt representational strategies qualitatively distinct from those of dense models.

Setup. We adopt the configuration of Section 4.1, augmented with the load-balancing term to prevent specialization collapse—the failure mode in which experts and router settle into a local minimum that leaves certain experts entirely unused—which arises in some configurations. We consider three configurations, each with a single active expert ($k = 1$): (A) $n = 2, m = 1$; (B) $n = 3, m = 1$; and (C) $n = 3, m = 2$. For each, we report the per-expert phase diagram over feature sparsity and last-feature relative importance, sweeping network sparsity by increasing the expert count E up to the number of input feature dimensions n .

In this section we hold active parameters fixed rather than total parameters, imposing $m_{dense} = km_{moe}$ (neglecting router parameters). This isolates architectural effects: fixing active parameters ensures that any observed difference is attributable to architecture rather than to a difference in active-parameter budget. Note that 4.C.1/1 has the same active-parameter count as 4.B.X/2; the latter, however, must encode the same number of input features ($n = 3$) with a single hidden dimension ($m = 1$) rather than two ($m = 2$), which precludes using superposition as a lens on specialization.

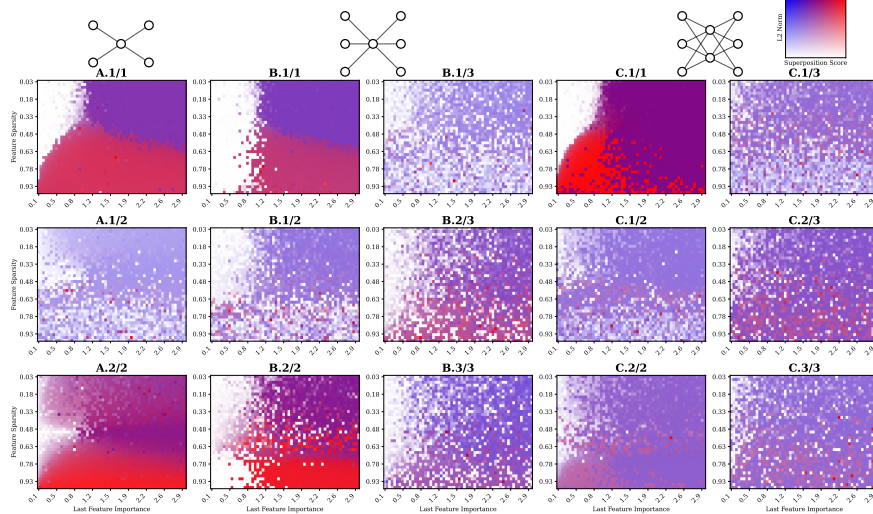


Figure 4: For a given expert and input dimension (feature), we decode how that feature is embedded in the hidden space—ignored (white), monosemantic (blue-purple), or in superposition (red). Each cell jointly encodes the feature norm ($\|W_n\|^2$) and the superposition score ($(\sum_{j < n} \hat{W}_n \cdot W_j)^2$) over feature sparsity $S \in [0.1, 1]$ and relative last-feature importance $I_n \in [0.1, 3]$, where the subscript n denotes the last of n total features. Per cell, we train ten models and retain the lowest-loss instance; the load-balancing term is active throughout this section. The color encoding is as follows: a low L_2 norm ($\|W_n\|$) yields white, indicating the model ignores the last feature; otherwise, a low superposition score yields blue-purple, indicating monosemantic representation, and a high score yields red, indicating superposition. Cell (i, j) in subfigure X.e/E denotes expert e of E total experts, trained on architecture X at last-feature importance $I_n = i$ and sparsity $S = j$; X.1/1 denotes the dense baseline.

Results and takeaways. In every single-expert (dense) configuration we observe a sharp phase transition (Figure 4.X.1/1), confirming Elhage et al. (2022). As the total expert count increases, the discrete phase transitions vanish. Some experts in the $E = 2$ MoEs resemble their dense counterparts (Figure 4.X.2/2) but exhibit smoother, more continuous transitions. In each case one expert becomes markedly more monosemantic, specializing in the feature of highest relative importance. Experts that diverge from the dense pattern uniformly

exhibit far lower superposition scores (i.e., appear bluer), indicating more monosemantic representations. This is consistent with the preceding section: MoEs systematically favor lower superposition than their dense counterparts.

In the $n = 2, m = 1$ configuration (Figure 4.A), the dense model fails to represent the last feature at low feature sparsity, whereas the matched MoE retains it far more reliably owing to its greater available capacity. With three input dimensions (Figure 4.B) this behavior does not recur, because the experts place the other two features in superposition, leaving no room for the third feature within a single hidden dimension. In architecture B, unlike the other two, superposition within one hidden dimension is insufficient to represent all features in the high-sparsity regime. Even so, no sharp phase transition emerges, except in the 0.5–0.7 feature-sparsity band of 4.B.2/2, where the boundary is largely discrete but mixed. For $m = 2$, the white region of the dense model (Figure 4.C.1/1)—in the mid- to low-sparsity domain, where the last feature is less important than the others—corresponds to the model ignoring that feature. As network sparsity increases (all remaining panels of Figure 4.C), the models represent the last feature with monotonically greater L_2 magnitude ($\|W_3^1\| < \|W_3^2\| < \|W_3^3\|$). Equivalently, the dimensionality allocated in the low-importance region increases with network sparsity, consistent with Figure 3.

Across Figures 4.B.2/2, 4.C.1/2, and 4.C.2/2 we observe a feature-sparsity window of roughly 0.48 to 0.7 in which polysemantic, monosemantic, and ignored representations are heavily intermixed. This indicates an intermediate regime in which MoEs incur comparable loss under either polysemantic or monosemantic representation, preventing consistent commitment to the strategies observed in the low- and high-sparsity domains. No analogous regime appears in dense models—further evidence that MoEs adopt distinct representational strategies.

Conclusion. All experiments in this section use top- $k = 1$ routing, so exactly one expert is active per input. Even under this restriction, the $E = 2$ case already exhibits qualitatively distinct behavior—including in configurations where superposition within the available hidden capacity would suffice to represent all features. We therefore conclude that treating a MoE as a mere aggregation of dense models is misleading: the routing mechanism, by exposing each expert to only a subset of the feature domain, substantially alters both the learned representations and the training dynamics of the experts.

6 EXPERT SPECIALIZATION

Having established that MoEs exhibit reduced superposition, we now analyze how the resulting monosemantic features are organized within experts and how this organization relates to specialization.

Expert specialization in MoEs is conventionally quantified through load balancing—the distribution of routed inputs across experts (Chaudhari et al., 2025). This criterion, however, does not capture the intuitive notion of specialization, under which an expert is invoked only when the concepts in which it specializes are present in the input.

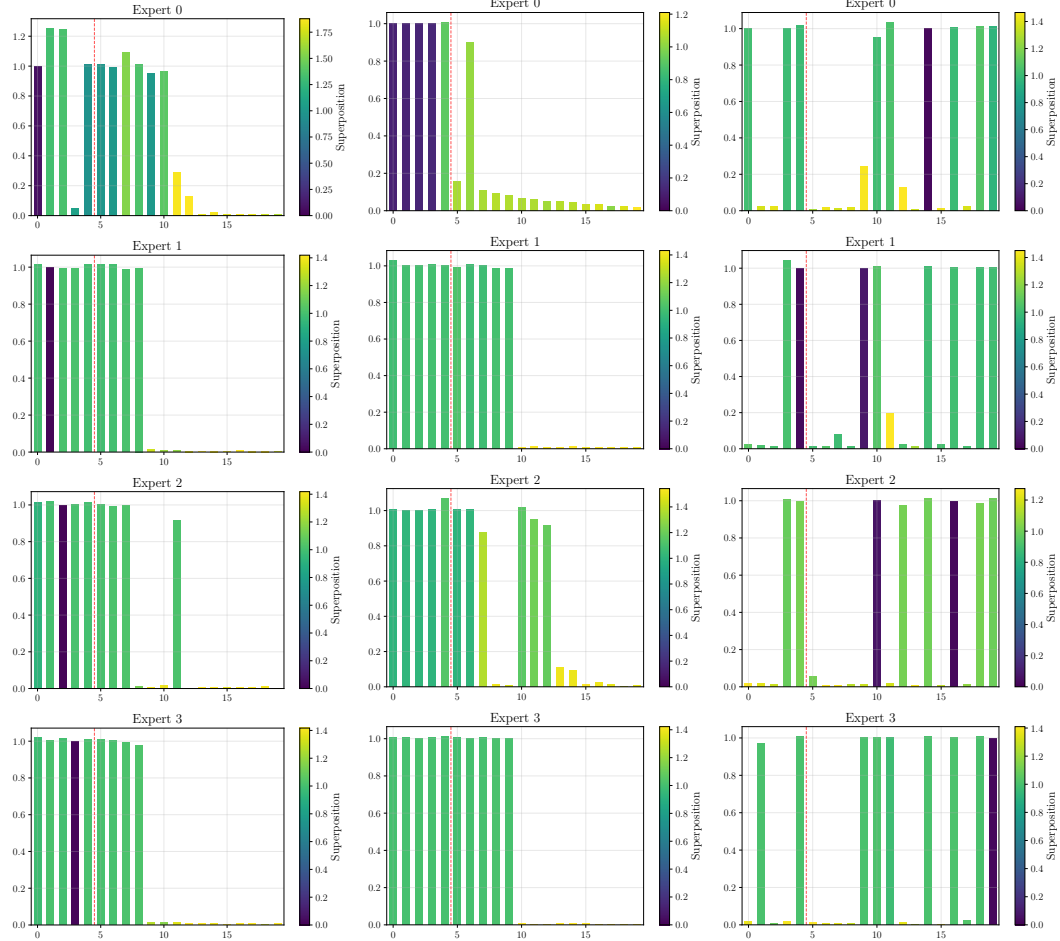
We define an expert to be specialized if it *occupies* a set of feature directions in the input space and represents those features relatively monosemantically. We show that these two conditions are directly correlated, and that their joint satisfaction promotes load balancing across experts.

Under the fixed choice $k = 1$, the router partitions the input space into convex cones (Appendix A.3), each of which is routed to a single expert. Formally, $\forall s > 0, \vec{x} \in C \rightarrow s\vec{x} \in C$, where C is the point set of the cone and s is any positive scalar. Consequently, if a feature vector $\vec{x}$ is routed to a given expert, then every positive scaling $s\vec{x}$ is routed to the same expert. In this case we say that feature x is contained within expert e , and equivalently that expert e *occupies* x .

We find empirically that small models which distribute the input space across a larger number of experts tend to achieve lower loss (Appendix A.4). This raises the question of whether the allocation of the input space to particular experts entails any representational

property of those experts. Our definition of specialization predicts that such allocation entails *monosemanticity*—a correlation that, as we show below, persists in larger toy models (e.g., $m = 10$).

For models with $n > 2$, we test two directions of this relationship. First, whether initializing an expert to occupy a set of input-space features induces monosemantic representation of those features. Second, and conversely, whether the features an expert represents monosemantically are also the features it occupies in the input space.



(a) Diagonal Initialization (b) Ordered K-hot initialization (c) Random K-hot initialization

Figure 5: Expert feature norms $\|W_i^{(e)}\|$ and superposition (color) under three router-initialization schemes, with $n = 20, m = 5, E = 4, S = 0.1$. In (a), the gate matrix is initialized on the main diagonal ($W_i^T = \hat{e}_i$, the standard basis vector for dimension i), with relative feature importance decreasing exponentially in index order from feature 1 to 20. In (b), the gate matrix is initialized as an “ordered k-hot” mapping, so that expert 1 aligns with the first five features and each subsequent expert aligns with the next five; the importance schedule matches (a). In (c), the gate matrix is initialized as a “random k-hot” mapping, in which each expert is assigned five features drawn so that experts share no common feature yet jointly cover all 20 features; importance again decreases exponentially but is permuted randomly across features.

Under diagonal initialization of the gate matrix, each expert monosemantically represents exactly the single feature it initially occupied, and no other (Figure 5a). Under ordered k-hot initialization, the first expert monosemantically represents four of the five features it initially occupied (Figure 5b); the remaining experts, initialized over lower-importance

features, represent neither those less important features nor the five most important features, which lay outside their initialization. When the importance ordering is broken and the features occupied by each expert are randomized, each expert monosemantically represents only the most important feature(s) within its initialization set (Figure 5c).

These results establish a strong correspondence between the features initially routed to an expert and the features that expert ultimately represents monosemantically. Moreover, experts represent only high-importance features monosemantically. This holds both when an expert is explicitly initialized with a single important feature and when it is initialized with a feature set, from which it autonomously selects the most important member for monosemantic representation.

Under uniform feature importance, experts instead place all features in superposition with elevated interference. Even so, the features an expert initially occupies remain *relatively* more monosemantic, attaining superposition scores on average 1.03 standard deviations below that expert’s mean.

We next test for a correlation between an expert’s monosemantic representation of a feature and its occupancy of that feature in the input. To this end, we measure expert usage under two conditions: when the feature is *one of several* active features, and when it is the *sole* active feature. The latter condition is equivalent to estimating the probability that the expert occupies the feature. The correlation holds under both Xavier and k-hot initialization (Table 1). For $E = 10$, a mean expert usage of $\sim 10\%$ corresponds to uniform load balancing across experts. In every case, activating an expert’s monosemantic feature(s) sharply increases that expert’s usage, and when such feature(s) constitute the only active input, the expert dominates routing. Under k-hot initialization, 100% of the features monosemantically represented by an expert are also occupied by that same expert.

Table 1: Per-expert monosemantic-feature and usage statistics for $n = 100, m = 10, E = 10$. For each initialization scheme (Xavier and k-hot) we train one hundred models, yielding 1000 experts per scheme. Statistics are aggregated across models with experts stratified by the number of features they represent monosemantically. For the feature(s) an expert represents monosemantically, we report the expert usage when those feature(s) are among several active input features, and the usage when they are the *only* active input feature.

Xavier Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage; feature(s) active (%)	Mean expert usage; only feature(s) active (%)
0	461	–	–	–
1	387	9.595	17.94	67.18
2	138	9.599	30.29	95.65
3	13	8.363	40.19	100.0
4	1	1.428	14.69	100.0
5	0	–	–	–
K-Hot Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage; feature(s) active (%)	Mean expert usage only feature(s) active (%)
0	335	–	–	–
1	382	10.00	23.94	100.0
2	227	10.02	46.61	100.0
3	47	10.09	62.00	100.0
4	8	9.95	70.30	100.0
5	1	9.62	74.79	100.0

The number of features an expert represents monosemantically thus serves as an index of its specialization. As this count grows, the expert’s unconditional usage on arbitrary inputs

decreases, whereas its usage conditional on its specialized features being active rises well above that of other experts. This pattern holds in all cases except the Xavier-initialized expert with four monosemantic features, for which unconditional utilization drops sharply.

7 CONCLUSION

We characterized how the expert decomposition modulates superposition in MoEs, establishing that MoEs consistently attain greater monosemanticity than dense networks while exhibiting no discrete phase transition. We further formalized a feature-based definition of expert specialization and demonstrated that experts organize around coherent feature subsets when the router initialization promotes this organization. These findings are obtained within simple autoencoder toy models on synthetic data, and their generalization to large-scale transformers with unknown feature distributions remains open (Appendix A.6). Subject to these constraints, we show that toy MoEs attain comparable loss while preserving more interpretable representations—contradicting the prevailing assumption that mechanistic interpretability and model capability are fundamentally in tension. Future work should identify the factors that favor monosemanticity in MoEs, characterize how MoE training dynamics diverge from those of dense models, and determine the conditions under which specialization emerges. Resolving these questions would inform the design of more interpretable, high-performing language models.

REFERENCES

- Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023. <https://transformer-circuits.pub/2023/monosemantic-features/index.html>.
- Marmik Chaudhari, Idhant Gulati, Nishkal Hundia, Pranav Karra, and Shivam Raval. Moe lens - an expert is all you need. In *Sparsity in LLMs (SLLM): Deep Dive into Mixture of Experts, Quantization, Hardware, and Inference*, 2025. URL <https://openreview.net/forum?id=GS4WXncwSF>.
- Michael B. Cohen, Yin Tat Lee, Cameron Musco, Christopher Musco, Richard Peng, and Aaron Sidford. Uniform sampling for matrix approximation. *CoRR*, abs/1408.5099, 2014. URL <http://arxiv.org/abs/1408.5099>.
- Nishanth Dikkala, Nikhil Ghosh, Raghu Meka, Rina Panigrahy, Nikhil Vyas, and Xin Wang. On the benefits of learning to route in mixture-of-experts models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 9376–9396, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.583. URL <https://aclanthology.org/2023.emnlp-main.583>.
- Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, Roger Grosse, Sam McCandlish, Jared Kaplan, Dario Amodei, Martin Wattenberg, and Christopher Olah. Toy models of superposition. *Transformer Circuits Pub*, 2022. URL https://transformer-circuits.pub/2022/toy_model/index.html.
- William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity, 2022. URL <https://arxiv.org/abs/2101.03961>.
- Google DeepMind. Gemini 2.5 pro. <https://deepmind.google/technologies/gemini/pro/>, 2025.

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- Wes Gurnee, Neel Nanda, Matthew Pauly, Katherine Harvey, Dmitrii Troitskii, and Dimitris Bertsimas. Finding neurons in a haystack: Case studies with sparse probing. *arXiv preprint arXiv:2305.01610*, 2023.
- Albert Q Jiang, Alexandre Sablayrolles, Antoine Roux, Arthur Mensch, Blanche Savary, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Emma Bou Hanna, Florian Bressand, et al. Mixtral of experts. *arXiv preprint arXiv:2401.04088*, 2024.
- Jakub Krajewski, Jan Ludziejewski, Kamil Adamczewski, Maciej Pióro, Michał Krutul, Szymon Antoniak, Kamil Ciebiera, Krystian Król, Tomasz Odrzygóźdź, Piotr Sankowski, Marek Cygan, and Sebastian Jaszczur. Scaling laws for fine-grained mixture of experts. *arXiv preprint arXiv:2402.07871*, 2024. URL <https://arxiv.org/abs/2402.07871>.
- Dmitry Lepikhin, HyoukJoong Lee, Yuanzhong Xu, Dehao Chen, Orhan Firat, Yanping Huang, Maxim Krikun, Noam Shazeer, Zhifeng Chen, Yonghui Wu, et al. Gshard: Scaling giant models with conditional computation and automatic sharding. In *Proceedings of the 38th International Conference on Machine Learning*. PMLR, 2021.
- Siyuan Mu and Sen Lin. A comprehensive survey of mixture-of-experts: Algorithms, theory, and applications, 03 2025.
- Jungwoo Park, Ahn Young Jin, Kee-Eung Kim, and Jaewoo Kang. Monet: Mixture of monosemantic experts for transformers. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=10gw1SHY3p>.
- Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models, 2024. URL <https://arxiv.org/abs/2311.03658>.
- Lucas Prieto, Edward Stevinson, Melih Barsbey, Tolga Birdal, and Pedro AM Mediano. Correlations in the data lead to semantically rich feature geometry under superposition. In *Mechanistic Interpretability Workshop at NeurIPS 2025*, 2025.
- Adam Scherlis, Kshitij Sachan, Adam S. Jermyn, Joe Benton, and Buck Shlegeris. Polysemanticity and capacity in neural networks, 2025. URL <https://arxiv.org/abs/2210.01892>.
- Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarczyk, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer, 2017. URL <https://arxiv.org/abs/1701.06538>.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025a.
- Xingyi Yang, Constantin Venhoff, Ashkan Khakzar, Christian Schroeder de Witt, Puneet K. Dokania, Adel Bibi, and Philip Torr. Mixture of experts made intrinsically interpretable. In *Forty-second International Conference on Machine Learning*, 2025b. URL <https://openreview.net/forum?id=6QERrXMLP2>.
- Yanqi Zhou, Tao Lei, Hanxiao Liu, Nan Du, Yanping Huang, Vincent Y Zhao, Andrew M. Dai, Zhifeng Chen, Quoc V Le, and James Laudon. Mixture-of-experts with expert choice routing. In Alice H. Oh, Alekh Agarwal, Danielle Belgrave, and Kyunghyun Cho (eds.), *Advances in Neural Information Processing Systems*, 2022. URL <https://openreview.net/forum?id=jdJo1HIVinI>.
- Barret Zoph, Irwan Bello, Sameer Kumar, Nan Du, Yanping Huang, Jeff Dean, Noam M. Shazeer, and William Fedus. St-moe: Designing stable and transferable sparse expert models, 2022. URL <https://api.semanticscholar.org/CorpusID:248496391>.

A APPENDIX

A.1 MEASURING LOSS FOR VARYING SPARSITY & EXPERTS

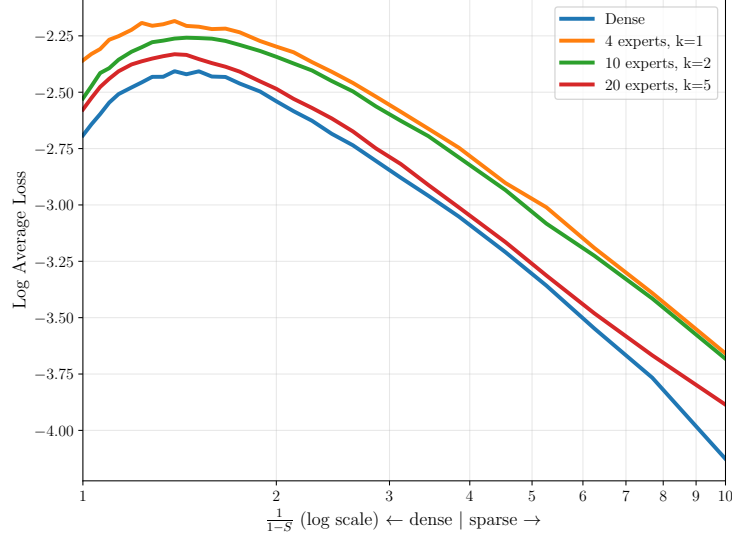


Figure 6: Log mean loss as a function of feature density ($\frac{1}{1-S}$) for the **dense** ($m = 20$) model and the **MoE** (4 experts, $k = 1$, $m = 5$), **MoE** (10 experts, $k = 2$, $m = 2$), and **MoE** (20 experts, $k = 5$, $m = 1$) configurations, all with uniform feature importance ($I_i = 1.0$) and $n = 100$ input features. Each point is averaged over five runs per sparsity level. The **dense** model attains the lowest loss at every sparsity level, but the gap to the **MoEs** contracts monotonically as the expert count increases.

A.2 THEORETICAL INTUITION FOR SUPERPOSITION

In dense models, superposition arises to exploit the mismatch between sparse features and dense computation, compressing rarely active features into shared dimensions. A MoE instead resolves this mismatch structurally, through conditional computation: by aligning activation sparsity with feature sparsity, it eliminates the computational penalty associated with dedicated, rarely activating neurons. As Elhage et al. (2022) observe, when a model expends computation only on active features, partitioning polysemantic neurons into dedicated monosemantic ones becomes optimal—effectively substituting the *selection* afforded by routing for the *compression* afforded by superposition.

Geometrically, this structural change manifests as a reduction in *interference*. Although the global ratio of represented features to parameters is unchanged, the router partitions the feature space so that a feature routed to expert e competes for capacity only with the subset of features co-assigned to e . Interference is therefore governed by the local per-expert Gram matrix $(W^e)^\top W^e$ rather than by the global $W^\top W$ of a dense model. This partitioning sharply reduces the number of interfering features for any given feature vector, attenuating the optimization pressure toward superposition and permitting experts to learn monosemantic representations.

A.3 ROUTER SUBSPACES ARE CONVEX CONES

In the $k = 1$ regime, the router $g(x) = \text{softmax}(W^r x)$ reduces to $\text{argmax}(W^r x)$. The region routed to expert i is the set $\{x : \forall j \neq i, (w_i - w_j)^\top x > 0\}$, where w_i and w_j are rows of W^r . Each constraint is a homogeneous linear inequality, and the intersection of such half-spaces is by definition a convex cone. Concretely, if x satisfies the inequality, so does sx

for any positive scalar s ; and if x_1 and x_2 both satisfy it, so does every convex combination $x = \lambda x_1 + (1 - \lambda)x_2$ with $\lambda \in [0, 1]$.

For $k > 1$, the region routed to a given expert e becomes a union of convex cones. Since a union of convex cones is not in general a convex cone, the interpretation of experts as occupying feature directions need not extend beyond $k = 1$.

A.4 EXPERT ROUTING WITH DIFFERENT INITIALIZATION SCHEMES

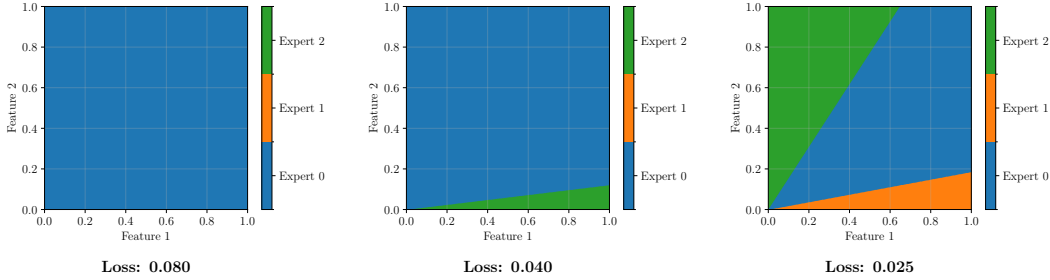


Figure 7: Expert routing for three architecturally identical models under distinct initialization schemes, with $n = 2$, $E = 3$, $m = 1$. The first model (left) attains the highest loss (0.08) and collapses all inputs onto a single expert. The second model (middle) attains lower loss (0.04) and routes a small subregion—inputs for which feature 1 is active—to a second expert. The third model (right) attains the lowest loss (0.025) and distributes the input space across all three experts: one is selected when only feature 1 is active, one when only feature 2 is active, and one when both are active.

For small models ($n = 2, m = 1, E > 1$), sweeping $E \in [2, 7]$, we find empirically that models distributing the input space across a larger number of experts attain lower loss. Fixing $n = 2$ permits direct visualization of which subregions of the input space are routed to which experts (Figure 7).

A.5 ANALYTIC MODEL EQUIVARIANCE

For the toy setting of single-layer, single-nonlinearity, top- $k = 1$ MoEs, we exhibit a theoretical mapping from any dense model to a monosemantic MoE that represents an equivalent number of active features under a sparsity constraint.

Assume an upper bound a on the number of active features per input, $\forall x \in D : |\{i : x_i \neq 0\}| \leq a$. Assume further that a does not exceed the per-expert hidden dimensionality m , which bounds the number of features any expert must represent, and that the hidden dimensionality is at most the total number of input features n , so that $a \leq m \leq n$. To construct the monosemantic MoE, instantiate, for each subset $S \subseteq \{1, 2, \dots, n\}$ with $|S| \leq a$ —that is, each admissible set of simultaneously active features—an expert that monosemantically preserves the features in S . (It in fact suffices to enumerate only the subsets with $|S| = a$.) The router then selects the expert corresponding to the active feature set, which by assumption never exceeds a features:

$$\text{Router}(x) = \arg \max_S \mathbb{I}[\text{support}(x) = S]$$

where $\text{support}(x) = \{i : x_i \neq 0\}$. Since $|S| \leq a \leq m$, each expert has capacity sufficient to represent its assigned features without superposition. In summary, at most a features are active at once, and every distinct active-feature combination is assigned a dedicated expert with adequate capacity for monosemantic representation, so the number of experts required is $\binom{n}{a}$.

This theoretical MoE achieves zero reconstruction loss only insofar as the sparsity constraint holds (equivalently, as sparsity approaches one in these toy models), because more than

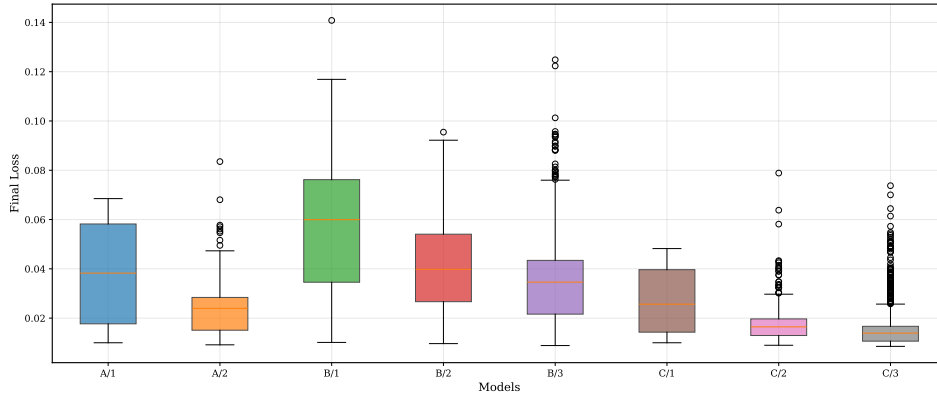


Figure 8: In the label X/E , X denotes the model architectures used in Figure 4 and E denotes the total expert count (i.e., network sparsity). Increasing network sparsity reduces mean loss while increasing its localized variance—most markedly as the expert count approaches the input feature dimensionality. We attribute the elevated variance to the comparatively unstable training of MoEs relative to dense models, which persists despite training ten models per cell and retaining the lowest-loss instance.

m features may occasionally be simultaneously active ($a \not\leq m$), exceeding the network’s monosemantic capacity. In that event, however, a dense polysemantic model fares no better unless the features are correlated in the distribution; hence, even when $a \leq m$ intermittently, the polysemantic model faces the same limitation, and the monosemantic MoE under this construction may still outperform it under looser sparsity constraints.

Thus, for any dense model $f_{\text{dense}}(x) = \text{ReLU}(Wx + b)$ subject to the sparsity constraint $|\text{support}(x)| \leq a$, there exists a MoE $f_{\text{MoE}}(x)$ satisfying $f_{\text{dense}}(x) = f_{\text{MoE}}(x)$ for all admissible inputs. In the toy settings of this paper, the sparsity constraint holds in the limit where sparsity approaches one. In practice, moreover, the number of features encodable within a given amount of information is likely bounded, as reflected by the dimensionality of meaningful embeddings of that data. Consequently, a MoE with sufficiently many experts and a tractable—i.e., interpretable—degree of superposition may suffice to encode all present features.

A.6 LIMITATIONS

Our results should be interpreted subject to several limitations. First, all experiments are conducted in controlled toy-model settings adapted from superposition studies on sparse autoencoders (Elhage et al., 2022). While this affords precise measurement of interference and monosemanticity, it abstracts away many properties of large-scale transformers—multi-layer computation, attention, heterogeneous feature distributions, and realistic routing dynamics—so the transferability of our findings to production MoE architectures remains uncertain.

Second, our architectural and routing choices are deliberately constrained: simple experts, fixed top- k routing (predominantly $k = 1$), equal parameter budgets across dense and MoE models, and no auxiliary load-balancing loss. Practical MoEs typically employ richer routing mechanisms, variable expert capacities, and multi-task objectives (Lepikhin et al., 2021; Fedus et al., 2022), or extend beyond reconstruction to next-token prediction, any of which may induce qualitatively different representational behavior.

Third, the feature distributions in our synthetic tasks—their sparsity patterns, importance schedules, and independence assumptions—are substantially simpler than those of natural data.

Finally, although we observe increased monosemanticity and reduced superposition in MoEs under fixed conditions, we do not evaluate downstream performance trade-offs, training

stability, or specialization dynamics at scale; prior work indicates that such dynamics can shift substantially with model size, optimization regime, and data diversity (Krajewski et al., 2024).

In summary, our study yields mechanistic insights under clean experimental conditions, but validating these patterns in large, realistic MoE systems requires further work. Notably, our methods do not scale naturally to larger models, which we identify as a clear direction for future work.